Database Design Documentation

1. Overview

This schema supports a manufacturing procurement control workflow for a Phase-1 MVP:

- Capture internal demand as Purchase Requests (PR)
- Route PRs through approval with auditability
- Generate RFQs from approved PR lines
- Track recipient vendors and quotation versions

Design philosophy

- **Control first:** all key business documents are persisted and status-driven.
- **Traceability by default:** PR/RFQ states, actors, and timestamps are retained.
- **Master-data-led transactions:** materials, vendors, units, and currencies drive document quality.
- **Document integrity:** important business rules are enforced at DB level (unique keys, checks, FKs).

2. Entity List (with business purpose)

Table	Business purpose
`departments`	Organizational units that own users and PRs.
`users`	Authenticated users with role-based responsibilities.
`password_reset_tokens`	Laravel password reset token store.
`sessions`	Laravel session persistence.
`jobs`	Async queue jobs.
`currencies`	Commercial currencies used in RFQ and vendor pricing.
`products`	Finished goods produced by the factory.

Table	Business purpose
`units`	Standard units of measure (UOM).
`raw_materials`	Procured materials required for production.
`bom`	Bill of materials mapping products to required materials and quantities.
`vendors`	Approved supplier master records.
`vendor_materials`	Vendor-material sourcing facts (price, currency, lead time, MOQ).
`pr_header`	PR document header (internal request).
`pr_lines`	Material line items within a PR.
`rfq`	RFQ document created from one approved PR line.
`rfq_recipients`	Vendor recipients attached to RFQ distribution.
`quotation`	Versioned vendor quotation responses per recipient.

3. Table Documentation

`departments`

Purpose: Department reference used by users and PR ownership.

Column	Type	Nullable	Description
`department_id`	`bigIncrements`	No	Primary key.
`name`	`string(100)`	No	Department name.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `departments (1) -> users (many)` via `users.department_id -> departments.department_id`
- `departments (1) -> pr_header (many)` via `pr_header.department_id -> departments.department_id`

Key constraints / rules

- Unique: `name`

`users`

Purpose: User identity, authentication attributes, and role assignment.

Column	Type	Nullable	Description
`user_id`	`bigIncrements`	No	Primary key.
`name`	`string(150)`	No	Full name.
`email`	`string`	No	Login email.
`email_verified_at`	`timestamp`	Yes	Verification timestamp.
`password`	`string`	No	Password hash.
`department_id`	`foreignId`	Yes	Optional department FK.
`role`	`enum`	No	`requester` , `procurement_manager` , `purchasing_officer` , `admin` .
`is_active`	`boolean`	No	Activation flag.
`remember_token`	`string`	Yes	Remember-me token.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `users (many) -> departments (1)` via `users.department_id -> departments.department_id`
- `users (1) -> pr_header (many)` via `pr_header.requester_id -> users.user_id`
- `users (1) -> pr_header (many)` via `pr_header.approved_by -> users.user_id`
- `users (1) -> pr_header (many)` via `pr_header.cancelled_by -> users.user_id`
- `users (1) -> rfq (many)` via `rfq.issued_by -> users.user_id`

Key constraints / rules

- Unique: `email`

- Index: `role, is_active`
- FK delete policy: `department_id` uses `ON DELETE SET NULL`

`sessions`

Purpose: Session store for authenticated web sessions.

Column	Type	Nullable	Description
`id`	`string`	No	Session primary key.
`user_id`	`foreignId`	Yes	Indexed user ID reference (not constrained by FK in migration).
`ip_address`	`string(45)`	Yes	Client IP.
`user_agent`	`text`	Yes	Browser user agent.
`payload`	`longText`	No	Serialized session payload.
`last_activity`	`integer`	No	Unix timestamp of last activity.

Relationships

- Logical reference from `sessions.user_id` to `users.user_id` (index only).

Key constraints / rules

- Primary key: `id`
- Indexes: `user_id`, `last_activity`

`currencies`

Purpose: Currency master used for RFQ commercial terms and vendor price context.

Column	Type	Nullable	Description
`id`	`bigIncrements`	No	Primary key.
`code`	`char(3)`	No	ISO-like currency code (e.g., USD).

Column	Type	Nullable	Description
`name`	`string(100)`	No	Currency display name.
`symbol`	`string(10)`	No	Symbol or short label.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `currencies (1) -> rfq (many)` via `rfq.currency_id -> currencies.id`
- `currencies (1) -> vendor_materials (many)` via `vendor_materials.currency_id -> currencies.id`

Key constraints / rules

- Unique: `code`

`products`

Purpose: Finished goods that define material demand through BOM.

Column	Type	Nullable	Description
`product_id`	`bigIncrements`	No	Primary key.
`code`	`string(30)`	No	Product code.
`name`	`string(150)`	No	Product name.
`description`	`text`	Yes	Optional description.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `products (1) -> bom (many)` via `bom.product_id -> products.product_id`

Key constraints / rules

- Unique: `code`

- Unique: `name`

`units`

Purpose: Reusable units of measure across master and transactional tables.

Column	Type	Nullable	Description
`unit_id`	`bigIncrements`	No	Primary key.
`name`	`string(50)`	No	Unit name.
`symbol`	`string(20)`	No	Unit symbol.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `units (1) -> raw_materials (many)` via `raw_materials.base_unit_id -> units.unit_id`
- `units (1) -> bom (many)` via `bom.unit_id -> units.unit_id`
- `units (1) -> pr_lines (many)` via `pr_lines.unit_id -> units.unit_id`
- `units (1) -> rfq (many)` via `rfq.unit_id -> units.unit_id`

Key constraints / rules

- Unique: `name`
- Unique: `symbol`

`raw_materials`

Purpose: Procurement items referenced in BOM, PR lines, RFQ snapshots, and sourcing.

Column	Type	Nullable	Description
`material_id`	`bigIncrements`	No	Primary key.
`name`	`string(150)`	No	Material name.

Column	Type	Nullable	Description
`base_unit_id`	`foreignId`	No	Base UOM FK.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `raw_materials (many) -> units (1)` via `raw_materials.base_unit_id -> units.unit_id`
- `raw_materials (1) -> bom (many)` via `bom.material_id -> raw_materials.material_id`
- `raw_materials (1) -> pr_lines (many)` via `pr_lines.material_id -> raw_materials.material_id`
- `raw_materials (1) -> rfq (many)` via `rfq.material_id -> raw_materials.material_id`
- `raw_materials (1) -> vendor_materials (many)` via `vendor_materials.material_id -> raw_materials.material_id`

Key constraints / rules

- Unique: `name`
- FK delete policy: `base_unit_id` uses `ON DELETE RESTRICT`

bom

Purpose: Product-to-material requirement lines (quantity and UOM).

Column	Type	Nullable	Description
`bom_id`	`bigIncrements`	No	Primary key.
`product_id`	`foreignId`	No	Product FK.
`material_id`	`foreignId`	No	Material FK.
`quantity_required`	`decimal(18,4)`	No	Required quantity per BOM line.
`unit_id`	`foreignId`	No	UOM FK.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `bom (many) -> products (1)` via `bom.product_id -> products.product_id`
- `bom (many) -> raw_materials (1)` via `bom.material_id -> raw_materials.material_id`
- `bom (many) -> units (1)` via `bom.unit_id -> units.unit_id`

Key constraints / rules

- Unique: `product_id, material_id`
- Check: `quantity_required > 0`
- FK delete policy: `product_id` cascade, `material_id` restrict, `unit_id` restrict

`vendors`

Purpose: Supplier master data used for sourcing and RFQ distribution.

Column	Type	Nullable	Description
`vendor_id`	`bigIncrements`	No	Primary key.
`name`	`string(150)`	No	Vendor legal/commercial name.
`email`	`string(255)`	No	Vendor contact email.
`phone`	`string(50)`	Yes	Vendor phone.
`country`	`string(100)`	Yes	Vendor country.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `vendors (1) -> vendor_materials (many)` via `vendor_materials.vendor_id -> vendors.vendor_id`
- `vendors (1) -> rfq_recipients (many)` via `rfq_recipients.vendor_id -> vendors.vendor_id`

Key constraints / rules

- Unique: `email`

`vendor_materials`

Purpose: Sourcing matrix between vendor and material with commercial and planning indicators.

Column	Type	Nullable	Description
`vendor_material_id`	`bigIncrements`	No	Primary key.
`vendor_id`	`foreignId`	No	Vendor FK.
`material_id`	`foreignId`	No	Material FK.
`lead_time_days`	`integer`	Yes	Typical lead time in days.
`minimum_order_qty`	`decimal(18,4)`	Yes	Minimum order quantity.
`preferred_vendor`	`boolean`	No	Preferred source flag.
`last_price`	`decimal(18,4)`	Yes	Last known price amount.
`currency_id`	`foreignId`	Yes	Currency for `last_price` .
`vendor_material_code`	`string(100)`	Yes	Vendor-side material identifier.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `vendor_materials (many) -> vendors (1)` via `vendor_materials.vendor_id -> vendors.vendor_id`
- `vendor_materials (many) -> raw_materials (1)` via `vendor_materials.material_id -> raw_materials.material_id`
- `vendor_materials (many) -> currencies (1)` via `vendor_materials.currency_id -> currencies.id`

Key constraints / rules

- Unique: `vendor_id, material_id`
- FK delete policy: `vendor_id` cascade, `material_id` cascade, `currency_id` restrict

`pr_header`

Purpose: Purchase Request document header with lifecycle and audit trail.

Column	Type	Nullable	Description
`pr_id`	`bigIncrements`	No	Primary key.
`pr_number`	`string(30)`	No	Generated document number (`PR-YYYYMMDD-XXXX`).
`requester_id`	`foreignId`	No	Requesting user FK.
`department_id`	`foreignId`	No	Department snapshot FK at creation time.
`request_date`	`date`	No	PR date.
`status`	`enum`	No	`draft` , `submitted` , `approved` , `cancelled` .
`notes`	`text`	Yes	Optional PR note.
`approved_by`	`foreignId`	Yes	Approving user FK.
`approved_at`	`timestamp`	Yes	Approval timestamp.
`cancelled_by`	`foreignId`	Yes	Cancelling user FK.
`cancelled_at`	`timestamp`	Yes	Cancellation timestamp.
`cancellation_reason`	`text`	Yes	Required at business layer when cancelling.
`deleted_at`	`timestamp`	Yes	Soft delete timestamp.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `pr_header (many) -> users (1)` via `requester_id -> users.user_id`
- `pr_header (many) -> departments (1)` via `department_id -> departments.department_id`
- `pr_header (many) -> users (1)` via `approved_by -> users.user_id`
- `pr_header (many) -> users (1)` via `cancelled_by -> users.user_id`
- `pr_header (1) -> pr_lines (many)` via `pr_lines.pr_id -> pr_header.pr_id`

Key constraints / rules

- Unique: `pr_number`

- Index: `status`
- Index: `requester_id`
- Soft deletes enabled
- FK delete policy: requester/department/approver/canceller are all `ON DELETE RESTRICT` (except nullable columns)

`pr_lines`

Purpose: Detailed material requests under a PR.

Column	Type	Nullable	Description
`pr_line_id`	`bigIncrements`	No	Primary key.
`pr_id`	`foreignId`	No	PR header FK.
`line_no`	`unsignedSmallInteger`	No	Sequence number inside PR.
`material_id`	`foreignId`	No	Requested material FK.
`quantity`	`decimal(18,4)`	No	Requested quantity.
`unit_id`	`foreignId`	No	Requested UOM FK.
`required_delivery_date`	`date`	No	Requested delivery date.
`notes`	`text`	Yes	Optional line notes.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `pr_lines (many) -> pr_header (1)` via `pr_id -> pr_header.pr_id`
- `pr_lines (many) -> raw_materials (1)` via `material_id -> raw_materials.material_id`
- `pr_lines (many) -> units (1)` via `unit_id -> units.unit_id`
- `pr_lines (1) -> rfq (0..1)` via `rfq.pr_line_id -> pr_lines.pr_line_id` (unique on RFQ side)

Key constraints / rules

- Unique: `pr_id, line_no`
- FK delete policy: `pr_id` cascade; `material_id` restrict; `unit_id` restrict

`rfq`

Purpose: Vendor-facing request document generated from one PR line.

Column	Type	Nullable	Description
`rfq_id`	`bigIncrements`	No	Primary key.
`pr_line_id`	`foreignId`	No	Source PR line FK (one RFQ per line).
`rfq_number`	`string(30)`	No	Generated document number (`RFQ-YYYYMMDD-XXXX`).
`material_id`	`foreignId`	No	Snapshot material FK copied from PR line.
`quantity`	`decimal(18,4)`	No	Snapshot quantity copied from PR line.
`unit_id`	`foreignId`	No	Snapshot unit FK copied from PR line.
`required_delivery_date`	`date`	No	Snapshot required date.
`rfq_date`	`date`	No	RFQ date.
`quotation_due_date`	`date`	No	Vendor quotation deadline.
`currency_id`	`foreignId`	Yes	Required business currency for responses (FK).
`payment_terms`	`string(255)`	Yes	Commercial payment terms.
`delivery_location`	`string(255)`	Yes	Delivery location text.
`pdf_path`	`string(500)`	Yes	Stored PDF path, if persisted.
`issued_by`	`foreignId`	Yes	Issuing user FK.
`issued_at`	`timestamp`	Yes	Issue timestamp.
`status`	`enum`	No	`draft` , `issued` , `closed` , `awarded` , `cancelled` .
`deleted_at`	`timestamp`	Yes	Soft delete timestamp.

Column	Type	Nullable	Description
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `rfq (many) -> pr_lines (1)` via `pr_line_id -> pr_lines.pr_line_id`
- `rfq (many) -> raw_materials (1)` via `material_id -> raw_materials.material_id`
- `rfq (many) -> units (1)` via `unit_id -> units.unit_id`
- `rfq (many) -> currencies (1)` via `currency_id -> currencies.id`
- `rfq (many) -> users (1)` via `issued_by -> users.user_id`
- `rfq (1) -> rfq_recipients (many)` via `rfq_recipients.rfq_id -> rfq.rfq_id`

Key constraints / rules

- Unique: `pr_line_id` (one RFQ per PR line)
- Unique: `rfq_number`
- Check: `quotation_due_date >= rfq_date`
- Check: `quantity > 0`
- Check: `status = 'draft' OR issued_by IS NOT NULL`
- Index: `status`
- Soft deletes enabled
- FK delete policy: source/snapshot refs are `ON DELETE RESTRICT` ; recipients cascade from RFQ

`rfq_recipients`

Purpose: Junction table defining which vendors receive each RFQ.

Column	Type	Nullable	Description
`recipient_id`	`bigIncrements`	No	Primary key.
`rfq_id`	`foreignId`	No	RFQ FK.
`vendor_id`	`foreignId`	No	Vendor FK.
`status`	`enum`	No	`pending` , `sent` , `responded` , `declined` , `expired` .

Column	Type	Nullable	Description
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `rfq_recipients (many) -> rfq (1)` via `rfq_id -> rfq.rfq_id`
- `rfq_recipients (many) -> vendors (1)` via `vendor_id -> vendors.vendor_id`
- `rfq_recipients (1) -> quotation (many)` via `quotation.recipient_id -> rfq_recipients.recipient_id`

Key constraints / rules

- Unique: `rfq_id, vendor_id` (same vendor cannot be added twice to one RFQ)
- FK delete policy: `rfq_id` cascade; `vendor_id` restrict

`quotation`

Purpose: Vendor quotations with revision/version control per recipient.

Column	Type	Nullable	Description
`quotation_id`	`bigIncrements`	No	Primary key.
`recipient_id`	`foreignId`	No	RFQ recipient FK.
`version_no`	`unsignedSmallInteger`	No	Sequential version per recipient.
`unit_price`	`decimal(18,4)`	No	Offered unit price.
`lead_time_days`	`integer`	Yes	Offered lead time in days.
`status`	`enum`	No	`draft`, `submitted`, `revised`, `withdrawn`, `rejected`, `awarded`.
`created_at`	`timestamp`	Yes	Creation timestamp.
`updated_at`	`timestamp`	Yes	Update timestamp.

Relationships

- `quotation (many) -> rfq_recipients (1)` via `recipient_id -> rfq_recipients.recipient_id`
- Currency is inherited through chain: `quotation -> rfq_recipients -> rfq -> currencies`

Key constraints / rules

- Unique: `recipient_id, version_no`
- Check: `unit_price > 0`
- Check: `lead_time_days IS NULL OR lead_time_days >= 0`
- Index: `status`
- FK delete policy: `recipient_id` cascade

4. ERD Summary (plain-English relationship chains)

1. **Demand chain:**
   A requester (user) creates a PR header, then adds one or more PR lines that reference materials and UOM.
2. **RFQ generation chain:**
   Each approved PR line can produce exactly one RFQ. The RFQ stores a snapshot of material, quantity, unit, and required date.
3. **Vendor distribution chain:**
   One RFQ is sent to many vendor recipients using `rfq_recipients` ; each recipient entry tracks delivery/response status.
4. **Quotation chain:**
   Each recipient can submit multiple quotation versions. Version numbering is unique per recipient.
5. **Commercial context chain:**
   RFQ header sets one currency (`rfq.currency_id`) for all quotes under that RFQ. Quotations do not store currency directly.
6. **Sourcing intelligence chain:**
   Vendor capability and latest commercial terms are modeled in `vendor_materials` including `last_price` , `currency_id` , lead time, and MOQ.

5. Design Decisions

Snapshot pattern

- `pr_header.department_id` is captured as a snapshot of requester's department at creation time.
- `rfq` captures snapshot fields from the source PR line (`material_id` , `quantity` , `unit_id` , `required_delivery_date`).
- Rationale: preserves historical correctness even if master data or PR line context changes later.

Soft deletes on business documents

- Enabled on `pr_header` and `rfq` .
- Rationale: cancellation/archival should not destroy audit history or reporting traceability.

Audit trail strategy

- PR approval/cancellation actor and timestamps are persisted (`approved_by` , `approved_at` , `cancelled_by` , `cancelled_at` , `cancellation_reason`).
- RFQ issuance actor and timestamp are persisted (`issued_by` , `issued_at`).
- Rationale: supports accountability and compliance-ready timeline reconstruction.

Currency architecture

- RFQ uses `currency_id` FK to define target commercial currency.
- Quotation table intentionally has **no currency field** and inherits currency from parent RFQ via recipient link.
- Vendor-material pricing uses `vendor_materials.currency_id` to preserve pricing context by supplier market.
- Rationale: one RFQ = one currency, reducing inconsistency across quotations for the same event.